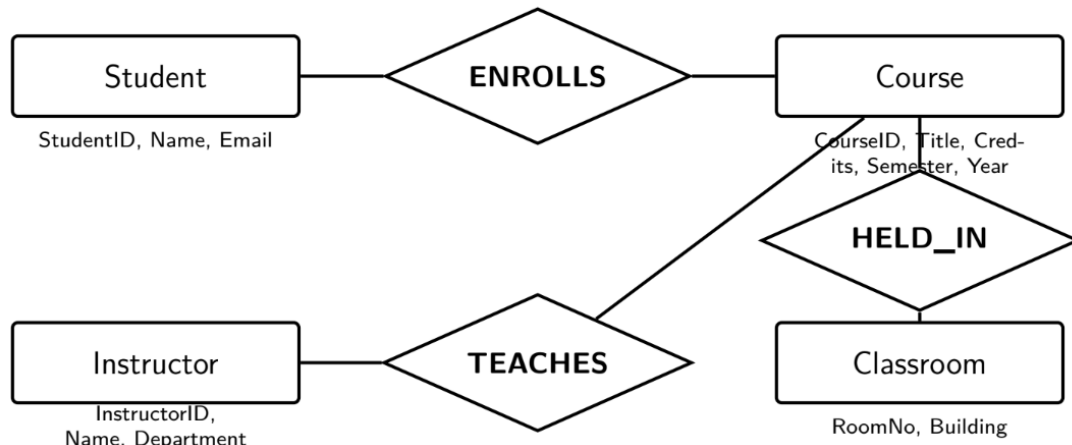


DBMS LAB ASSIGNMENT – 1

1. Given ER diagram-

Question 1 — University Course Registration



Study the diagram first. Business rules are on the next slide.

Rules-

Question 1 — Business Rules & Task

Business rules

- ▶ The same course is offered in many semesters and years.
- ▶ A course may have multiple sections in the same semester.
- ▶ Different sections may be taught by different instructors.
- ▶ A student registers for a particular section, not merely for a course.
- ▶ Classroom allocation may differ across sections.

Find exactly 4 issues.

For each issue:

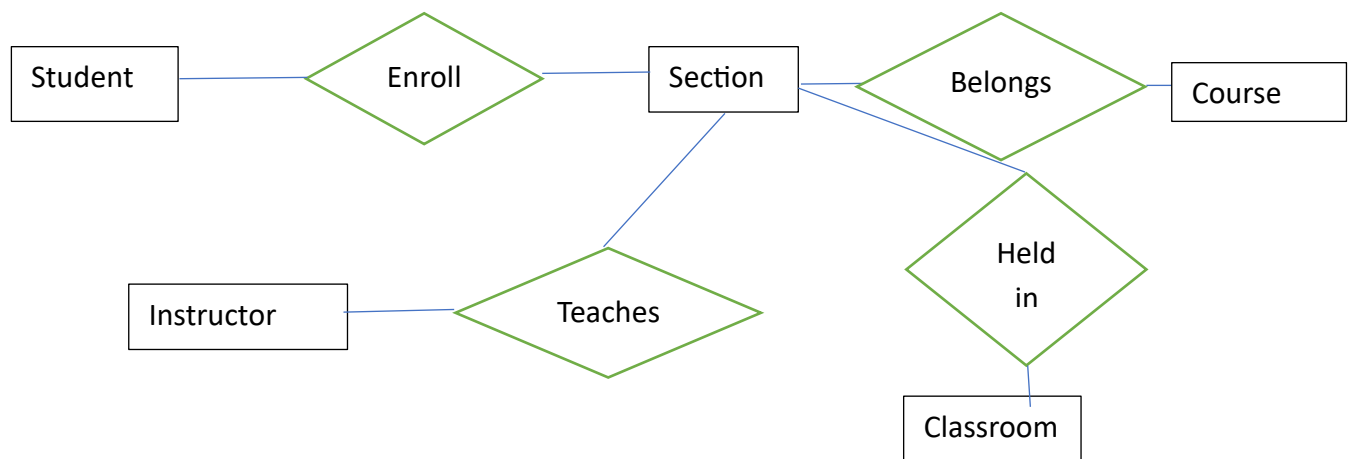
1. Identify the flaw.
2. Explain the information loss.
3. Give the minimal ER correction.

Flaws:

As given in the rules that a course can have different sections and each course has a separate instructor. Hence, there should be an entity named "section" and student enrolls in the section and each course has sections.

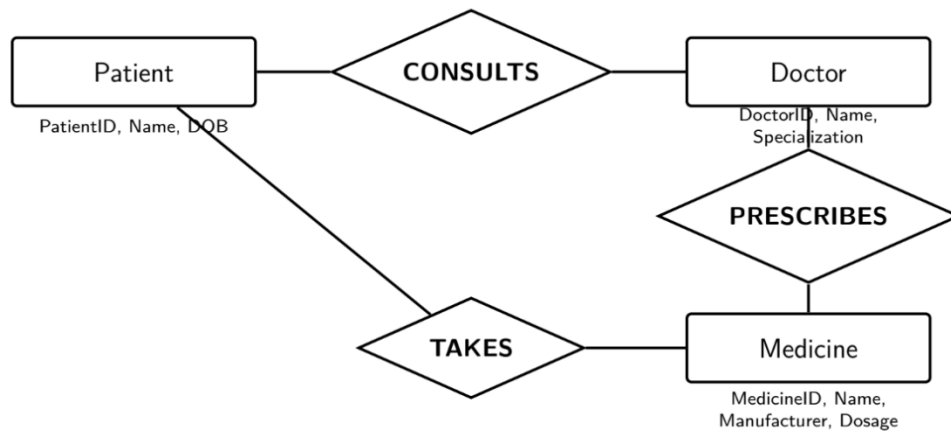
1. Semester and Year are attributes of Section as the same course is offered in many sections and years and if they are the attributes of Course , we might lost the data of the different semesters and years the course is offered.
2. Student should enroll in section and not in Course as the student registers for a section and not a course.
3. Since different sections of the Course can be taught by different instructors. Hence , Instructor teaches a section and not a Course.
4. Classroom allocation may differ among sections and hence a particular section is held in a classroom and not a Course.

Correct ER diagram :



2. Given ER diagram-

Question 2 — Hospital Prescription System



Study the diagram first. Business rules are on the next slide.

Rules-

Question 2 — Business Rules & Task

Business rules

- ▶ A patient may consult the same doctor many times.
- ▶ Different consultations may result in different prescriptions.
- ▶ The dosage of the same medicine may differ by patient and visit.
- ▶ The same medicine may be prescribed again with a different frequency or duration.
- ▶ A prescription must identify **who prescribed what, to whom, and when**.

Find exactly 4 issues.

Your redesign must preserve:

- repeated consultations,
- prescription history,
- patient-specific dosage,
- time-dependent instructions.

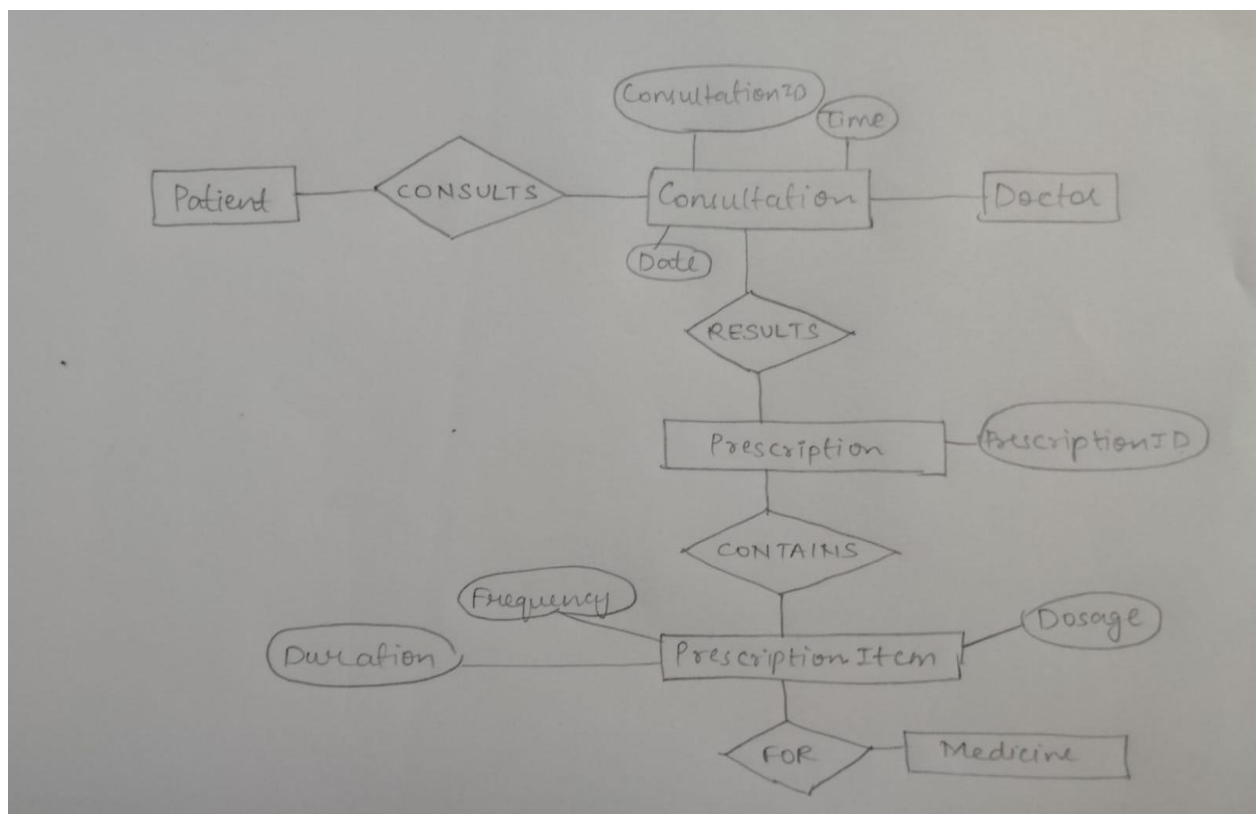
Flaws:

Since we need to store the consultation history and a patient might consult the same doctor many times, we need an entity "consultation", also one consultation results in "prescription" and that prescription will have certain "prescriptionItem" because the dosage is the attribute of prescriptionItem not the medicine.

1. Consultation should be an entity -> Patient ---CONSULTS---Consultation-----Doctor
2. Prescription should be an entity as well to keep the history whenever a patient consults a doctor.

3. PrescriptionItem is an entity which stores dosage as same medicine can be given in different dosages to two different individuals.
4. For time dependent instructions, we can put Frequency and Duration as attributes for PrescriptionItem.

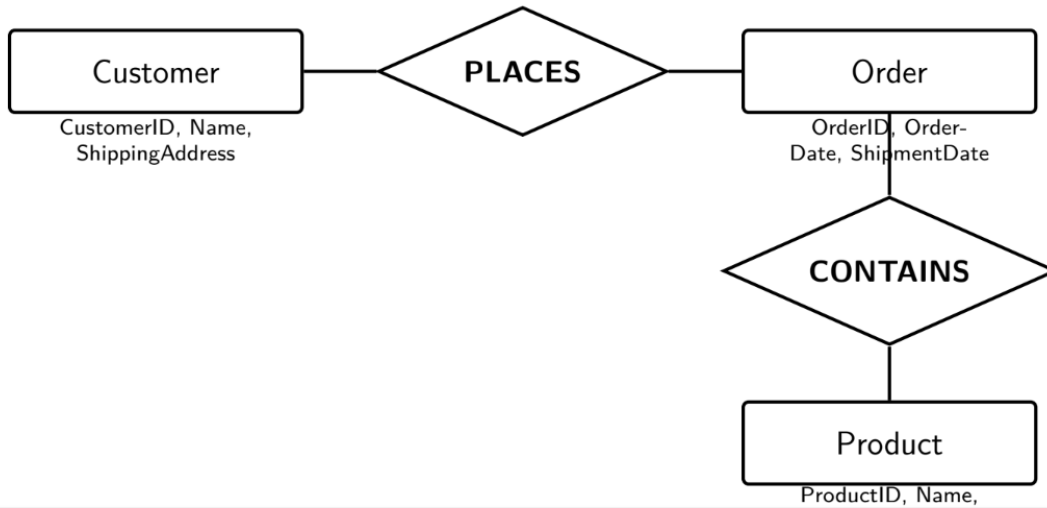
Correct ER diagram :



3. Given ER diagram-

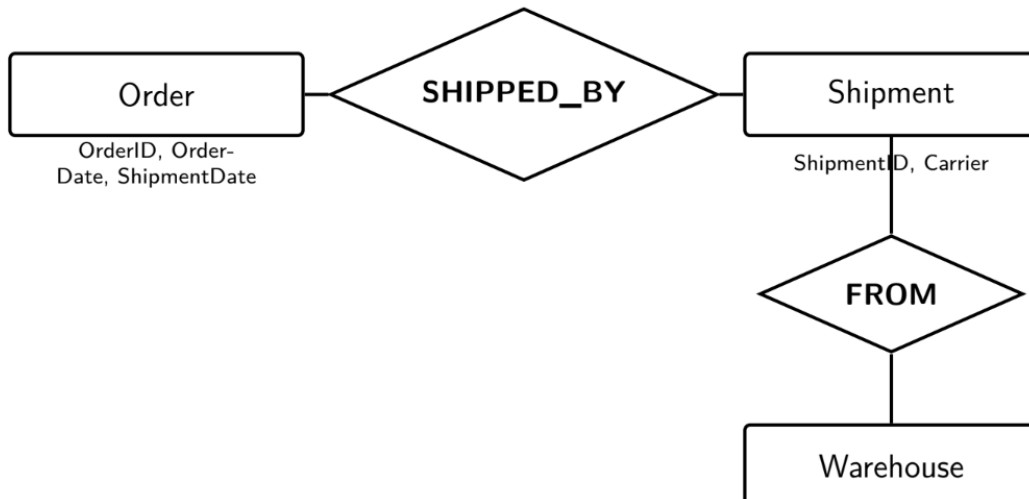
Question 3 — E-Commerce Order Fulfilment

Flawed ER Design — Part A



Question 3 — E-Commerce Order Fulfilment

Flawed ER Design — Part B



Rules-

Question 3 — Business Rules & Task

Business rules

- ▶ A customer may use a different delivery address for every order.
- ▶ Quantity is specific to an order line, not to the product itself.
- ▶ One order may be split into several shipments.
- ▶ A single order line may be partially shipped from different warehouses.
- ▶ Each shipment has its own dispatch date and carrier information.

Find exactly 4 issues.

The corrected model must support:

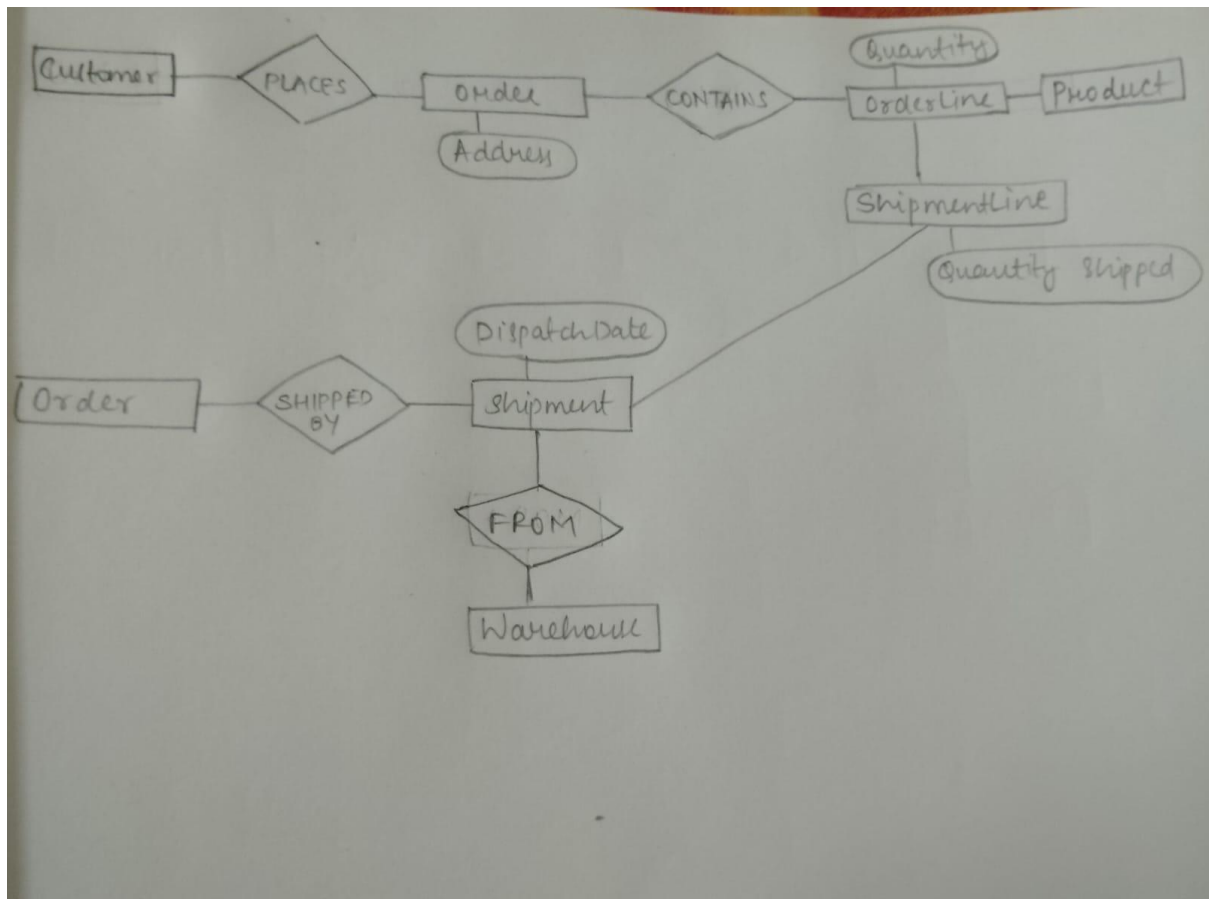
- historical order data,
- line-item quantity,
- split shipments,
- partial fulfilment.

Flaws:

The “OrderLine” entity is missing and is required as quantity is specific to that orderline and not the product. Also other entity “ShipmentLine” is required for splitting the shipment.

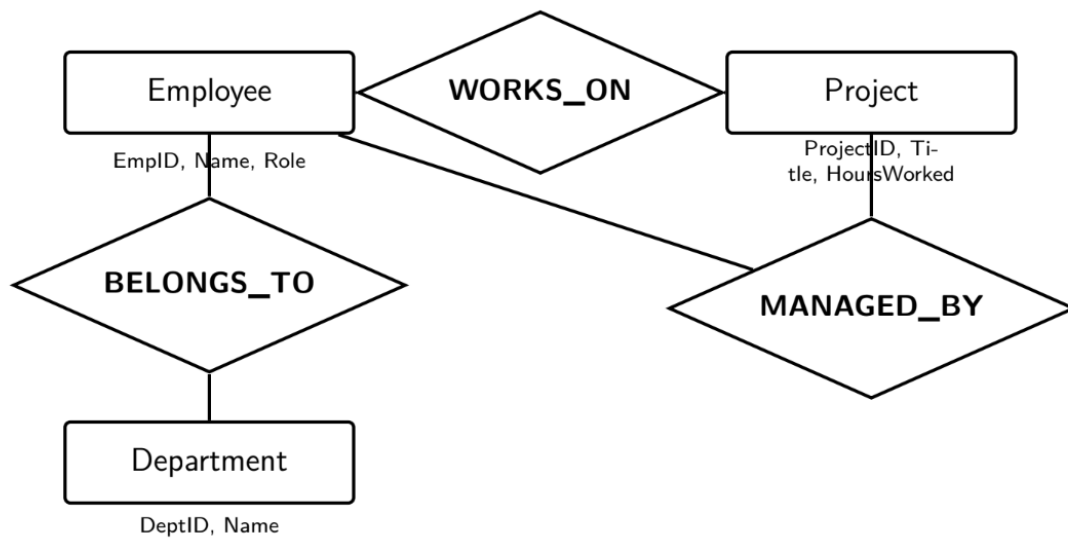
1. Address should be an entity for Order and not the Customer as the customer might use different address for two different orders.
2. OrderLine entity is required as the quantity is entitled to the OrderLine . So , Order----CONTAINS-----OrderLine-----Product
3. ShipmentLine entity is required for keeping the track of patially shipped order line.
4. Keep Shipment with its own Dispatch date so that multiple shipments can be allowed per order

Correct ER diagram:



4. Given ER diagram-

Question 4 — Project Staffing and Roles



Rules-

Question 4 — Business Rules & Task

Business rules

- ▶ An employee may join, leave, and later rejoin the same project.
- ▶ The employee's role may be different on different projects.
- ▶ Hours are recorded per employee-project assignment and time period.
- ▶ Project managers may change during the lifetime of a project.
- ▶ Management history must be retained for audit purposes.

Find exactly 4 issues.

Focus on information that is:

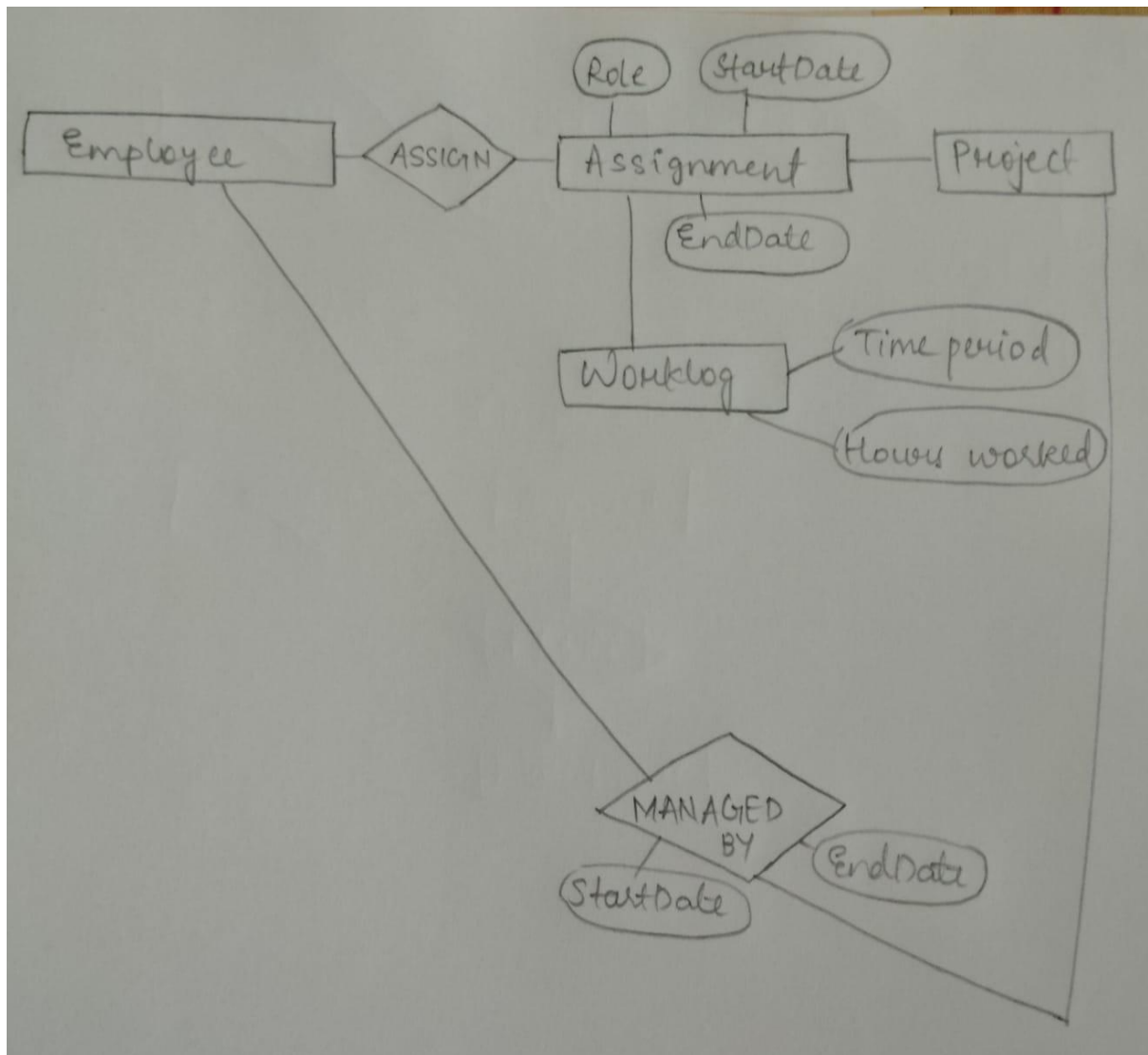
- assignment-specific,
- time-dependent, or
- historical.

Flaws:

The relationship between Employee and Project should also be made an entity . By doing this , we would be able to manage the Employee-Project history.

1. There should be an Assignment entity which stores the role of that particular employee in a project.
2. Again , the Works_on relationship cannot represent joining , leaving and rejoining . So , Assignment should be entitled with Start and End date.
3. Since hours are recorded per employee-project assignment. So , it can't be stored in the Project entity neither in Assignment because It requires hours per time period , Hence, we can make a new entity WorkLogs which stores hours worked per time period per assignment.
4. We should keep Start and End date as attributes for the MANAGED_BY relationship as managers might change over a period of time.

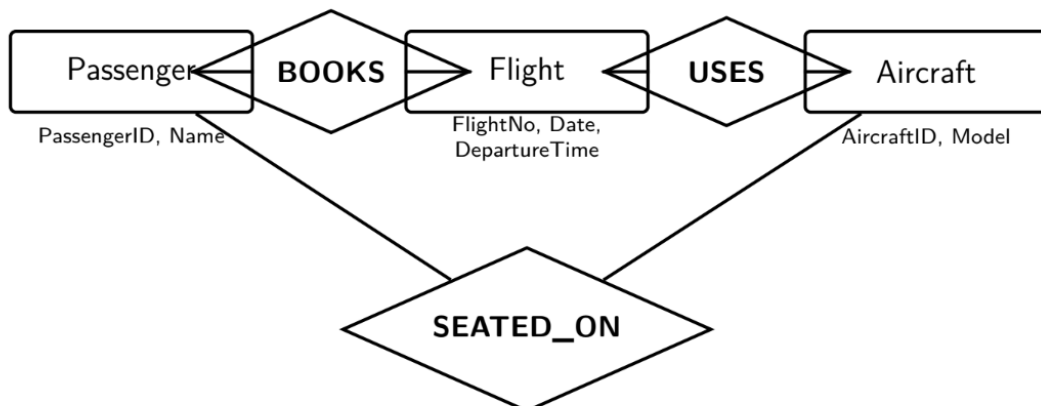
Correct ER diagram:



5. Given ER diagram-

Question 5 — Airline Booking and Seat Assignment

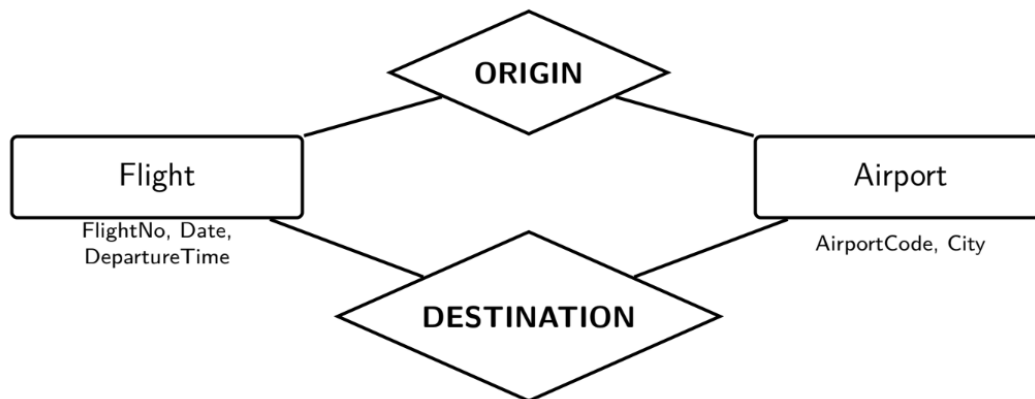
Flawed ER Design — Part A



Part B of the same ER model is on the next slide.

Question 5 — Airline Booking and Seat Assignment

Flawed ER Design — Part B



Parts A and B together form **one flawed ER model**.

Rules-

Question 5 — Business Rules & Task

Business rules

- ▶ A flight number operates on many dates.
- ▶ One PNR may contain several passengers and several flight segments.
- ▶ Aircraft assignment may change for one particular flight occurrence.
- ▶ A seat is assigned to a passenger for a specific flight occurrence.
- ▶ The same passenger may have different seats on different segments.

Find exactly 4 issues.

The corrected model must represent:

- PNR/bookings,
- flight occurrences,
- aircraft changes,
- segment-specific seats.

Flaws:

The current ER diagram does not properly represent bookings, flight occurrences, aircraft changes, and seat assignments.

1. There should be a PNR entity because one PNR can contain multiple passengers and multiple flight segments.

2. There should be a FlightOccurrence entity because same flight number can operate on different dates.
3. Aircraft entity should be connected to FlightOccurrence because the aircraft assigned to a flight may change for different occurrences.
4. SeatAssignment entity should be there to store different seats for the same passenger on different flight segments.

Correct ER diagram:

